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The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2022

COURSE CODE: NR 240

COURSE NAME: SOIL SCIENCE

TIME ALLOWED: THREE (3) HOURS

DATE: 21ST OCTOBER 2022

INSTRUCTIONS: 1. Answer the COMPULSORY question one in section A

2. Answer any FOUR questions from section B

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SECTION A (COMPULSORY QUESTION)

Question 1

a) Mr. John Mwape collected two soil samples from his Bentley Farm in the interior of Chinsali District and requested for a soil texture analysis. Briefly describe 3 methods of determining soil texture.

(6 marks)

b) The two soil samples had size separates determined by one of the methods in (a) above. The following results were obtained.

	Soil A	**Soil B**
Coarse fragments (2-50 mm diameter) =	20 g	35 g
Sand (2- 0.05 mm diameter) =	16 g	10 g
Silt (0.05-0.002 mm diameter) =	28 g	25 g
Clay (less than 0.002 mm diameter) =	12 g	150 g
Total mass	**220 g**	**220 g**

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(i) Using the textural triangle provided in *Figure 1*, on the appendix, determine the textural class names for the two soils. ***(6 marks)***

(ii). Discuss the differences in the soil properties of soils A and B in relation to chemical reactivity and nutrient availability, erosion potential and soil workability. ***(6 marks)***

(iii) Soil consistency describes the resistance of a soil at various moisture contents to mechanical stress or manipulation. Explain why it is important to determine soil consistency. ***(2 marks)***

SECTION B

Question 2

Most of Zambia's plateau and some hilly areas have been degraded due to geomorphological processes. Zambia has three main agro-ecological zones that comprise of different soil types. In brief outline the short falls and strength's associated with the soils of the three regions. ***(20 marks)***

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Question 3

The prudent use of soils depends on recognition of soil properties and predictions of soil behavior under various conditions. Using the Munsell system, colour is described in terms of three (3) properties.

(a) State and define the 3 properties used to describe soil colour using the Munsell system. *(6 marks)*

(b) Give an example of a Munsell notation clearly indicating the properties stated in (a) above. *(4 marks)*

(c) What is the significance of knowing the structure of the soil for Agroforestry scholars? *(4 marks)*

(d) What are the major characteristics of the **A** and **E** horizons of the soil profile? *(6 marks)*

Question 4

Distinguish the main features of physical and chemical weathering in soil formation. *(20 marks)*

Question 5

A soil has a CEC of 15 cmol (+)/kg. Calculate the amount of exchangeable calcium (Ca^{2+}) it retain in kg/ha if the soil has a depth of 20 cm and a bulk density of 1300 kg/m^3? *(20 marks)*

Question 6

(i). Define and give *two* ways on how the water content in the soil can be expressed? *(4 marks)*

(ii) Calculate the ***bulk density***, ***porosity*** and ***volumetric water content*** of a cylindrical kaolinitic soil sample with dimensions 12 cm long and 7.7cm in diameter that is 17% moisture content and weighs 688 g. *(16 marks)*

Question 7

(i) Calculate the porosity of a 252 g soil sample that contains 65 g of water when 53% of the pores are full of water. *(15 marks)*

(ii) What is the interpretation of Soil Bulk Density values in Soil Health? *(5 marks)*

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APPENDICES

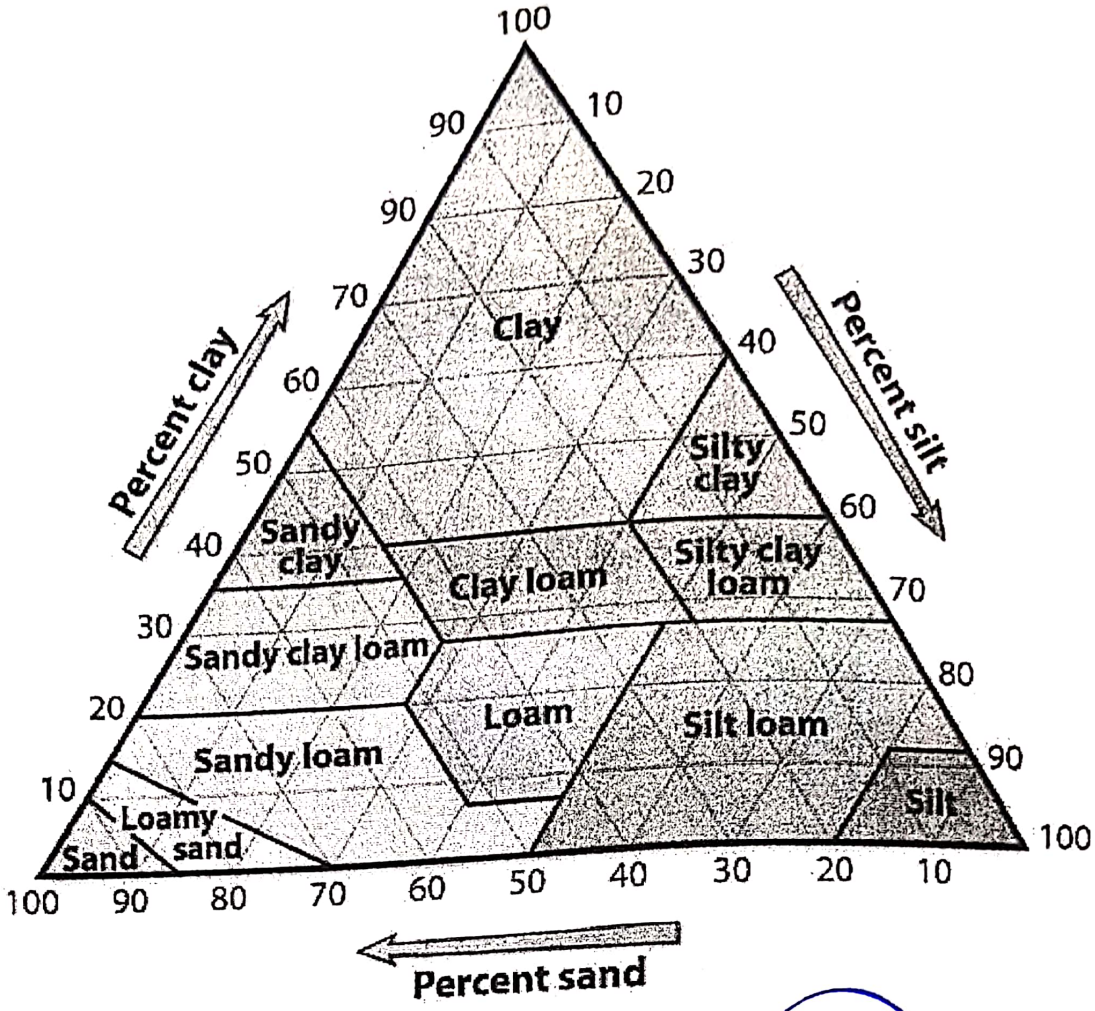


Fig 1: Soil Textural Triangle.

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The Copperbelt University
Kapasa Makasa Campus
Department of Agriculture and Aquatic Sciences

NR 240/ FO 260 - Soil Science

Date: 9 November 2021

Time: 14:00 – 17: 00

INSTRUCTIONS

Section A – Answer all Questions

Section B – Answer any two Questions

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SECTION A – Answer all Questions. Be brief and Concise with the answer

1. A farmer took a soil sample from his field for a soil test. When he got the results of the soil test, he realized that the **Cation Exchangce Capacity** (CEC) was very low. As an agronomist, mention three things the farmer can do to improve the CEC of this soil (**3 Marks**).

2. The soils results of a soil test showed that the soil had 80% Base saturation, 15% Exchangeable sodium percentage (ESP), 12 meq/100g Calcium, 2 meq/100g Magnesium and 4 meq/100g potassium. Using this information, calculate the **Cation Exchange Capacity** (CEC) of this soil? (**5 Marks**)

3. A farmer went into the field to determine the **soil bulk density**. Using a core ring with dimensions of 5 cm height and 5 cm diameter and a weight of 0.5g, she collected a soil sample from a field that was **saturated** with water. She then put the **soil sample together with the core ring** in the oven for 24 hours. After removing the sample from the oven, she discovered that the combined weight of the soil and the core ring had dropped by 5 % from the weight she recorded when she took the sample in the field. She calculated the bulk density and found a value of 1.5 g/cm^3. Using this information, calculate:

   (i) the porosity or pore percentage of this soil (**2.5 Marks**).
   (ii) the weight of the wet soil (**2.5 Marks**)
   (iii) the weight of the oven dry soil (**2.5 Marks**).

4. Explain what you understand by **Matric potential** and **Osmotic potential** (**4 Marks**)

5. Explain what you understand by **Adhesion** forces and **Cohesion** forces. (**4 Marks**)

6. A farmer discovers that his tomato crops are wilting despite giving the crop sufficient water. Explain a three possible Soil related problems that can cause this situation (**3 Marks**)

7. Describe how flocculation and Dispersion occurs in soils. (**3 Marks**)

8. Explain the role that micro-organisms play in improving crop production in soils? (**3 Marks**)

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9. Outline the key characteristics or properties of the following soil types (**6 marks**).
    a. Lixisols
    b. Acrisols
    c. Alfisols

10. Given the following full soil name; **Typic Hapludult,** clearly identify the soil classification levels according to Soil Taxonomy System (USDA) (**4 marks**).

11. Outline the major structural similarities and differences between mica muscovite and mica phlogopite (**4 Marks**)

SECTION B – Answer any two (2) of the three (3) questions given. Each question carries 7.5 Marks

Molar Mass: Ca- 40, C-12, O- 16, S= 32, P=31

1. Mr. Mudenda's soil test results show that his farm has a bulk density of 1300 Kg/m^3 and the calcium levels are 2 mg Ca/Kg soil. He intends to plant tomato in this field and from literature, he discovers that the total demand for Calcium by tomato is 200 Kg/Ha. Using a soil depth of 20 cm, calculate how much Gypsum ($CaSO_4$) in Kg/Ha) Mr. Mudenda needs to add to this soil in order to meet the Calcium tomato crop demand. (**7.5 Marks**)

2. A soil is found to have an Exchangeable Sodium percentage (ESP) of 17 % and a CEC of 32 meq/100g. Madalitso wants to reduce the Exchangeable Sodium Percentage (ESP) to 7% by applying a Calcium carbonate ($CaCO_3$) to the soil. Using a soil depth of 20 cm and a bulk density of 1300Kg/m^3, calculate how much $CaCO_3$ (in Kg/Ha) he will need to add to the soil in order to achieve this (**7.5 Marks**)

3. After realizing that the cost of chemical (inorganic) fertilizers has become unaffordable, a farmer decides to use a green manure as an alternative source of nutrients to meet crop requirements. How many kilograms of this green manure does this farmer need to fertilize a 1 ha field with a rate of 25 kg P/ha? Assume that the green manure can supply 2.25 g P/kg of green manure (**7.5 Marks**).

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KAPASA MAKASA CAMPUS

DEPARTMENT OF AGRICULTURE AND AQUATIC SCIENCES

2019 SESSIONBAL EXAMINATIONS

(NR 240/ FO 260)

SOIL SCIENCE

DATE: 1st OCTOBER 2019

TIME: 14:00 – 17: 00

INSTRUCTIONS

1. Answer all questions in Section A
2. Answer one of the two Questions in Section B
3. Show all your working
4. Time allocated for the exam is 3 Hours

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SECTION A – ANSWER ALL QUESTIONS (45 MARKS)

1. What do you understand by the term Regolith? **(2 marks)**

2. What is the **E horizon** of a soil profile? **(2 marks)**

3. What is the difference between **Felsic** and **Mafic** Igneous rocks? **(2 marks)**

4. Draw the structure of **Kaolinite** showing the tetrahedra and octahedra sheet and the different ions that are part of the structure? **(3 marks)**

5. What is the average cation exchange capacity **(CEC)** of Kaolinite and Organic matter? **(3 marks)**

6. What do you understand by the calcium carbonate equivalence **(CCE)** of a liming material? **(2 marks)**

7. What is **soil salinity**? How do you determine soil salinity of a soil? **(3 marks)**

8. What is the difference between **Mineralization** and **Humification? (3 marks)**

9. What is the difference between **Humus** and **organic matter**? **(3 marks)**

10. Mention three differences that exist between **Humic acids** and **fulvic acids? (6 marks)**

11. What is **field water capacity? (2 marks)**

12. In which soil moisture condition is the total **soil water potential** equals **zero? (2 marks)**

13. What is the difference between **crop water requirements** and **Irrigation requirement? (3 marks)**

14. What is **soil structure**? Name four types of soil structure. **(5 marks)**

15. Derive the formula $\mathbf{e = \phi/(1-\phi)}$ using the following equations : $\phi = (V_w + V_a)/V_t$ *and* $e = (V_w + V_a)/V_s$

Where

V_s = *volume of solids*

V_a = *Volume of air*

V_t = *Total soil volume*

e = *void ratio*

ϕ = *soil porosity* **(4 marks)**

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SECTION B- CHOOSE ANY ONE OF THE TWO QUESTIONS (15 MARKS)

Mass numbers

Ca = 40, S= 30, Na = 23, K = 39, Mg = 24, O = 16, H= 1. Use P_s = 2.65 g/cm³, P_w = 1g/cm³

1.A soil was found to have the following properties: Cation exchange capacity (CEC) of 3.2 meq/100g, Calcium (Ca^{2+}) was 0.82 meq/100g soil, Magnesium (Mg^{2+}) was 0.11 meq/100g soil and potassium (K) was 0.25meq/100g soil and exchangeable ($Al^{3+} + H^{+}$) of 0.03 meq/100g. Calculate:

a) The exchangeable sodium percentage (ESP) **(3 marks)**

b) The Base saturation of this soil? **(5 marks**

c) Did this soil have a problem of sodicity? **(2 marks)**

d) The amount of $CaSO_4.2H_2O$ (Kg/ha) required to neutralize ESP from the current level to 2%. (Take bulk density to be 1,7g/cm³ and soil depth to be 20 cm) **(5 marks)**

2. A student goes into the field after a heavy downpour of rain. He finds the soil is saturated with water and inserts a **core-ring** to take a soil sample. The **core-ring** has a diameter of 5 cm and a height of 4 cm. He carefully removes the **core-ring** from the soil and takes the sample to the laboratory. He weighs the **core-ring + soil** and records the results. He places the core-ring in the oven for 24 hours after which he realizes the **core-ring + soil** has lost a weight of 8g. Using this information, calculate: (Note for all calculations, take particle density as 2.65g/cm³ and water density 1g/cm³)

a) Bulk density **(3 marks)**

b) Gravimetric water content **(4 marks)**

c) Volumetric water content **(2 marks)**

d) The volume of soil particles only **(2 marks)**

e) In order to reach a gravimetric water content of 0.1 cm³ H_2O/cm³ soil, what is the volume of H_2O per Hectare of this soil that must be lost? (Take a depth of 20 cm) **(4 marks)**

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10.5/25 (11/25)

KAPASA MAKASA UNIVERSITY

COURSE: SOIL AND WATER MANAGEMENT –SA340 QUIZ #1

DURATION: 20 MINUTES

NAME: CATHERINE CHABUYE SIN No. 22171991 DATE: 15/02/2024

Question No.1

Define the following terms;

(i) Ped *(2 marks)*
(ii) Cation *(2 marks)*
(iii) Clastic *(2 marks)*
(iv) Gneiss *(2 marks)*

Question No.2

Indicate True (T) or False (F) *(10 marks)*

(i) Clay particles have a poor drainage ( ) T ✓(1)
(ii) The atmosphere contains mainly inert gases such as argon and helium ( ) F ✓(1)
(iii) Petrogenesis is the study of Rocks ( ) F ✓(1)
(iv) The Lithosphere refers to a part of the earth predominantly composed of solid rocky. ( ) T ✓(1)
(v) The biosphere mainly consists of the crust and the mantle ( ) F ✓(1)
(vi) Continental crust is uniform in thickness and composition. ( ) F ✓(1)
(vii) The feel method in particle size analysis give estimates of the clay content.( ) T X F
(viii) Soil stickiness' is attributed to particles < 0.002mm ( ) T ✓(1)
(ix) Kaolinite a 1:1 Clay tend to exhibit the expanding and shrinking characteristics. ( ) T X F
(x) Pumice is an extrusive fine grained igneous rock. ( ) F ✓(1)

Question No. 3

List the Soil forming processes (4 marks)

(i) Climate
(ii) Topography
(iii) Time
(iv) Parent Material

X(0)

- Correction
- Grain addition
- Losses
- Transformation
- Translocation

Question No.4

Give two examples of rocks that can be described as below, *(3 marks)*

1.Igneous Rocks... magma and Basalt ✓(0.5) - granite
2.Sedimentary Rocks... Gattine and Basalt ?✓(0.5) - limestone
3.Metamorphic Rock... Marble and Basalt ?✓(0.5) - Gneiss

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COURSE: SOIL AND WATER MANAGEMENT –SA340 TEST 1

DURATION: 1.5 HRS

TEST 1

Question No.1

Define the following terms;

(i) Laminar structure *(2 marks)*
(ii) Plutonic Rocks *(2 marks)*
(iii) Peat Soil *(2 marks)*
(iv) Young Soil *(2 marks)*
(v) Eluviation *(2 marks)*

Question No.2

Phosphorus and sulphur are plant nutrients. Do the levels of nutrient elements such as sulphur and phosphorus in the earth's crust justify the need for fertilizer (nutrient containing substances) in plant production? Explain. *(5 marks)*

Question No. 3

Define the following terms, (from i and ii)

(i) Air-filled Porosity *(2 marks)*
(ii) Volume wetness *(2 marks)*
(iii) Calculate the porosity of a 250 g sample that contains 65 g of water when 55% of the pores are full of water. *(8 marks)*
(iv) Calculate the bulk density of a soil sample that has a porosity of 45%. *(10 marks)*

Question No.4

Describe the roles of the ***Dokuchaeiv's*** soil forming factors in the formation of soils and their interrelationships. *(15 marks)*

Question No.5

Illustrate using the colloidal soil complex mechanism the cause of soil acidity in high rainfall areas in the northern part of Zambia and highlight the major limitation as regards to crop production? *(8 marks)*

Question No.6

A soil has a pH (0.01 M CaCl2) of 4.22, what is the agronomic interpretation of the pH of this soil? *(5marks)*

Question No.7

In modern day agriculture, the use of farm power mechanization is one of the conduits in boosting the production and productivity of crops. Tractor mechanization in tilling agriculture land possess a negative effective to soil health. Justify the assertion? *(5 marks)*

-END-

$$f = \frac{V_w}{V_a + V_f}$$

$$f_a = F - S$$

TEST 2

KAPASA MAKASA UNIVERSITY

COURSE: SOIL AND WATER MANAGEMENT – SA340 TEST II

DURATION: 1.5 HRS

INSTRUCTIONS: Answer all questions and be precise.

Question No.1

Define the following terms;

(i) Nitrification ***(2 marks)***
(ii) Flocculation ***(2 marks)***
(iii) Ion Adsorption ***(2 marks)***
(iv) Deprotonation ***(2 marks)***
(v) P^H Dependent Charge ***(2 marks***

Question No.2.

A soil has a CEC of 15 cmol (+)/kg. Calculate the amount of exchangeable calcium (Ca2+) it retain in kg/ha if the soil has a depth of 20 cm and a bulk density of 1300 kg/m^3? ***(15 marks)***

Question No. 3

Define the following terms, (from i to iii)

(i) Cation Exchange Capacity (CEC) ***(2 marks)***
(ii) Isormophous Substitution ***(2 marks)***
(iii) E-Horizon ***(2 marks)***
(iv) A soil from KMU contains 37 % clay and has a CEC of 1.72 cmol (+)/kg soil. Calculate the CEC of the clay fraction assuming the measured CEC is due to clay only. ***(10marks)***

Question No.4

The prudent use of soils depends on recognition of soil properties and predictions of soil behavior under various conditions.

a. Using the Munsell system, color is described in terms of three (3) properties.

(i) State and define the 3 properties used to describe soil color using the Munsell system. (***6marks***)

(ii) Give an example of a Munsell notation clearly indicating the properties stated in (i) above. (***4marks***)

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COURSE: SOIL AND WATER MANAGEMENT -SA340 TEST 3

DURATION: 1HR 15 MINUTES

***INSTRUCTIONS**: Answer all questions and be precise.*

Question No.1

Define the following terms;

(i) Darcy's Law *(2 marks)*
(ii) Infiltration Rate *(2 marks)*
(iii) Intensity of Irrigation *(2 marks)*
(iv) Field Capacity *(2 marks)*
(v) Hydraulic Conductivity *(2 marks)*

$\psi_s = RCT$

$R = 0.08314$
$C = M$
$T = °C$

Question No.2.

(i) Compare and contrast the ***Matric Potential*** Vs ***Osmotic Potential***. **(6 marks)**

(ii) A plant tissue was placed in an open beaker of 0.3M sucrose at 24°C. What is the water potential of the solution in the beaker? *(7 marks)*

(iii) If a plant tissue above has a water potential of -9 bars. Which way will the water move? How will this affect the cell? *(7 marks)*

Question No. 3

A field sample of an unconfined aquifer is packed in a cylinder (with length 50cm and diameter 6cm), the field sample is tested for 3 minutes under constant head difference of 16. 3cm.Then 45.2cm^3 is collected at the outlet. Determine the hydraulic conductivity of the aquifer sample? *(15 marks)*

Question No.4

Gravitational, Capillary and hydroscopic are three classes of moisture in the soil. Discuss the classes in relation to soil properties? *(10 marks)*

Question No.5

Briefly explain the major differences between drip irrigation and other irrigation systems. *(10 marks)*

- END OF TEST-

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COURSE: SOIL AND WATER MANAGEMENT –SA340 QUIZ #2

DURATION: 15 MINUTES

llowing terms;

(ii) desorption *(2 marks)*
(ii) Suction *(2 marks)*
(iii) Bar *(2 marks)*
(iv) Dipole *(2 marks)*

Question No.2

Indicate True (T) or False (F) *(10 marks)*

(i) Centrifugal force is key in gravitation water movement. (F)
(ii) A decrease in Surface tension increases the capillary water. (T)
(iii) Water at field capacity is readily unavailable to plants and some fungi. (F)
(iv) Saturated Water flow is possible at a water potential of more than -3.3 Kpa. (T)
(v) Kaolinite is a typical clay where Hysteresis process occurs. (F)
(vi) Darcy's law governs the Saturated Flow of Soil water movements. (T)
(vii) Soil water is a nutrient itself for Plant growth. (T)
(viii) Surface Tension is important in the Infiltration of Soil water. (T)
(ix) Permeability indicates a state when adhesion forces are at Zero. (F)
(x) Percolation is important in ground water recharge in wells meant for Irrigation. (T)

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soil

A root placed in a beaker of pure water has $\Psi_s = -0.15$ MPa (mega pascals) 1 MPa = 10 bars. In which direction will the net flow of water be.

The value of Ψ in a root tissue was found to be -3.3 bars. If you take the root tissue and place it in a 0.1 molar solution of sucrose at 20°C in an open beaker. What is the potential (ψ) value of the solution and which direction will the net flow of water be.